SAE prog_(RT1SA05) -- Data Processing Connection statistics of a web server

Abstract

The proposed SAÉ consists in writing a program in C language allowing to generate statistics on the connections of a web server. The program will have three types of interfaces: by default, an interface for the web and, depending on the call options, a graphical interface and a text interface.

You will have to:

1. Write a program to process log files of an apache server offering three interface modes (two graphical modes allowing to visualize the statistics in the form of histograms and a text mode).
2. Set up a demonstration model to test the program in real conditions of use.

Tools

To realize this SAÉ you will work under Linux using exclusively the virtual machine (VM) available on eCampus.

If you have not already done so, install VirtualBox or VMWare on your host machine. The provided virtual machine is small enough to fit on a 16GB USB stick. Download it, copy it to a directory on your host machine where you have write access.

Look for the VM on eCampus (either on the SAE dedicated space RT1SA05 or on the one corresponding to the ressource RT1R07), file `VirtualDebian13.ova` for VirtualBox, `VirtualDebian13_OVF.zip` for VMWare Workstation or VMWare Fusion on an Intel Macintosh, `VirtualDebian13_ARM_OVF.zip` if you are running an ARM architecture (typically on Apple Silicon based Macintosh).

The Apache web server is already installed and configured to accept CGI. The Apache configurations are standard: the root of the site is `/var/www/html/`, the CGI directory is `/usr/lib/cgi-bin/` (to write access to these two directories you will have to use the sudo command).

In the same location on eCampus, you will find the files `RT1SA05-2025.tgz` and `RT1SA05-2025-ARM.tgz`. Download the one corresponding to your architecture from the virtual machine, retrieve this file (it is usually downloaded in the ~/Téléchargement directory) and place it in /home/etudiant. Then, open a command terminal and do:

```
etudiant@virtualdebian:~$ sudo apt install git
etudiant@virtualdebian:~$ git clone http://github.com/ehuot/RT1SA05
etudiant@virtualdebian:~$ cd RT1SA05/
etudiant@virtualdebian:~/RT1SA05$ make install
etudiant@virtualdebian:~/RT1SA05$ firefox "http://127.0.0.1"
```

Thus, you are ready to work. A sample code can be found in the directory `/home/etudiant/RT1SA05/src/` but the `Makefile` to compile and install the program is not provided.

Example of a realization

The figure below visualizes the execution of the program `/usr/lib/cgi-bin/versiondynamique` (you do not have the source code of this program). Here, it is used in its default version (*ie.* the web interface):

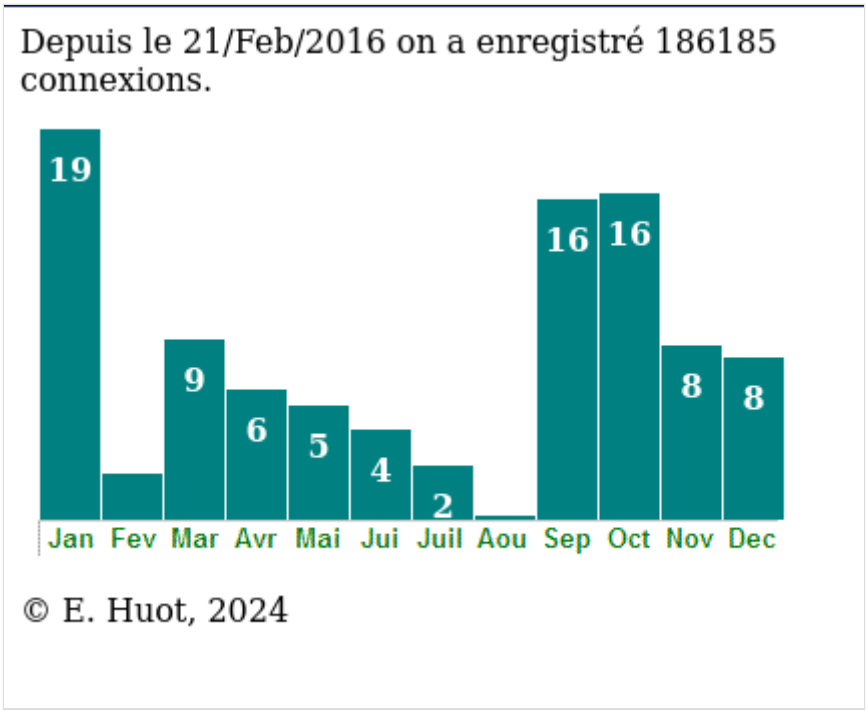


Figure 1: Connection statistics (web display).

The figure below shows a screen shot of the same program, using its graphical interface, while it is launched with its `-gr` option. Note that you are not asked to view this image *via* the web interface:

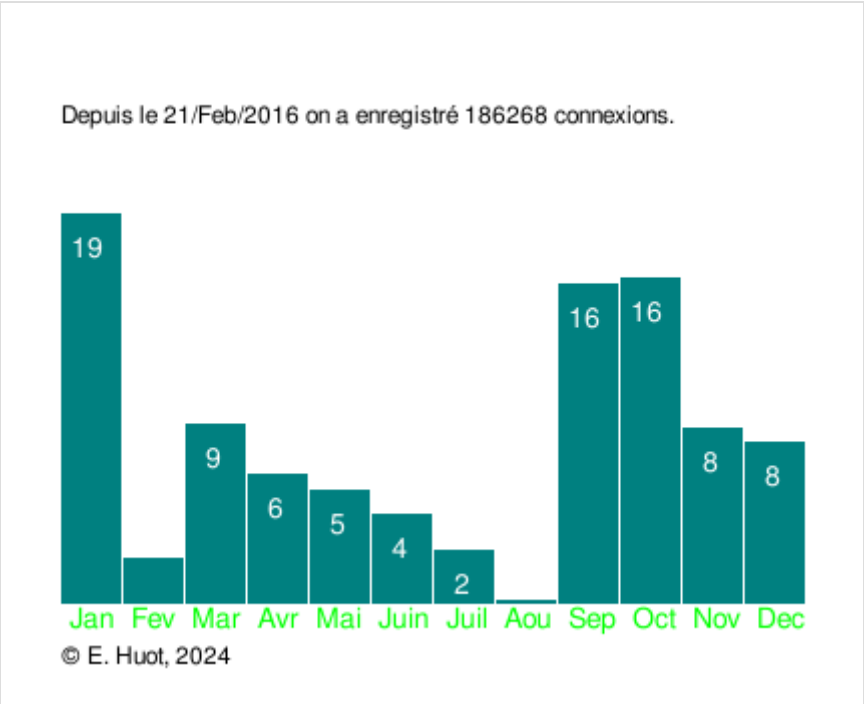


Figure 2: Connection statistics (graphic display).

The figure below shows a screen shot of the same program, using its textual interface, while it is launched with its `-txt` option. Note that you are not asked to view this image *via* the web interface:

```
Depuis le 21/Feb/2016 on a enregistre 186184
connexions.
   Janvier : 19.5%
   Fevrier :  2.3%
      Mars :  9.1%
     Avril :  6.6%
       Mai :  5.7%
      Juin :  4.5%
   Juillet :  2.7%
      Aout :  0.2%
 Septembre : 16.1%
   Octobre : 16.3%
  Novembre :  8.8%
  Decembre :  8.1%

(c) E. Huot, 2024
```

Figure 3: Connection statistics (text display).

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Subject

Context

You are a web master and system administrator, and you are asked to create a program that allows you to make some simple statistical calculations on the connections to this site. The results of these calculations must be able to be visualized either *via* a display in text mode, or *via* a graphical interface proposing a representation in histogram, or via a specialized page of the site. You have to write a program, in C language, performing the calculations based on the file `/var/log/apache2/access.log` summarizing the trace of all the connections to the site and proposing the three user interfaces.

Todo tasks

Your program must necessarily analyze the contents of the file `/var/log/apache2/access.log` and visualize:

- The total amount of requests recorded since the date of the first connection (each line of the log file corresponds to one request).
- A monthly distribution of these requests.

Bonus: you can propose other types of statistics (for example: the three most frequently used browsers).

Your program must necessarily offer three operating modes corresponding to the three types of interfaces listed below:

web (default interface).
: The program behaves like a CGI. Installed in the `/usr/lib/cgi-bin` directory. It generates HTML code and allows to visualize the monthly distribution of requests in the form of a histogram.

graphic (`-gr` option).
: The program opens a graphical window and displays the information. The monthly distribution of requests is visualized in the form of a histogram.

text (`-txt` option).
: The program displays the results as text in the terminal.

Work to be done and submitted

You must provide a Proof of Concept (PoC) of your website, containing of course your program (compilable and executable) as well as all the files necessary for its use. All your files will be compressed in a `.tgz` archive file (see submission).

This archive file will contain:

1. In the `RT1SA05/compterendu` directory, your report in PDF format (a .pdf, no other format accepted). This file will include the algorithms (you will explain why you chose this or that

data structure and present your algorithms in French in a simple pseudo-language), the listings of your source codes (**YES** it is mandatory to put your source codes both in the report and – of course – in the `src` directory), examples of execution traces (screen shots).
2. The HTML and CSS files needed to visualize the web interface.
3. All your `.c` and `.h` source files that will be placed in the `RT1SA05/src/` directory.
4. A `Makefile` allowing to compile (by a call to the `make` command) and to install (`make install`) your program in `/usr/lib/cgi-bin/` and the `HTML/CSS` files in `/var/www/html/`.

Tip list

The subject is not difficult but it is long and has some deliberately technical points. I do advice you to:

1. Start working on a sheet of paper and analyzing the `access.log` file by yourself.
2. Decompose your program into subprograms using top-down analysis.
3. Define the data structure(s) to be used at each step.
4. Test each step independently.

Keep it simple before you make it complex. Do not submit a program where you have thought of everything but that does not work. Prefer to propose a program that works but does not have all the functionalities you have been asked for.

Notation

Everything that is requested is evaluated (the filenames, their location, the respect of the submission guidelines, the presence of a `Makefile` and its behaviour, the functionalities, *etc*).

Approximately the grade will be split into three large and equal parts: theory (mainly algorithmic), program and other.

Failure to comply with an instruction resulting in the impossibility of correcting all or part of your work will result in a zero.

To evaluate your work we will start from the virtual machine that has been provided, download your last submission, type the command sequence:

```
etudiant@virtualdebian:~$ tar xvf binome1binome2.tgz
etudiant@virtualdebian:~$ cd RT1SA05/
etudiant@virtualdebian:~/RT1SA05$ make
etudiant@virtualdebian:~/RT1SA05$ make install
etudiant@virtualdebian:~/RT1SA05$ cd src/
etudiant@virtualdebian:~/RT1SA05/src$ ./versiondynamique -txt
etudiant@virtualdebian:~/RT1SA05/src$ ./versiondynamique -gr
etudiant@virtualdebian:~/RT1SA05/src$ cd ..
etudiant@virtualdebian:~/RT1SA05$ evince compterendu/*.pdf
etudiant@virtualdebian:~/RT1SA05$ firefox "http://127.0.0.1"
```

and check that everything is consistent and works properly.

Every topic is evaluated. We place great importance on:

- explaining your choices and the relevance of data structures and algorithms;
- breaking the program down into functions and subfunctions (avoid any redundancy);
- structuring the code (multiple .h and .c files);
- demonstrating a solid understanding of parameter passing;
- etc.

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Logs

Logging

To manage correctly a web server, it is necessary to have a feedback about the activity and performance of the server, as well as any problems that may arise. Most web servers allow you to keep track of requests and possibly problems that occur while using them. This is commonly referred as "log files" or, more commonly, *logs*.

The Apache server is very flexible and allows the administrator to configure the *logs*.

Common Log Format

The usual configuration will save log entries in a format known as Common Log Format (CLF). Log file entries generated in CLF format look like this:

```
1157.55.39.254 - - [01/Dec/2018:00:42:25 +0100] "GET /index.php HTTP/1.1"
302 495 "-" "Mozilla/5.0 (iPhone; CPU iPhone OS 7_0 like Mac OS X)
AppleWebKit/537.51.1"
```

Each part of this log entry is described in the following:

`1157.55.39.254`
: This is the IP address (or the name) of the client (the remote host) that sent the request to the server.

`-`
: Hyphens (*ie* minus characters) indicate that the corresponding portion of information is not available.

`[01/Dec/2018:00:42:25 +0100]`
: The date and time the request was received. The format is as follow:

```
[day/month/year:hour:minutes:secondes zone]
day = 2 digits
month = 3 letters
year = 4 digits
hour = 2 digits
minutes = 2 digits
secondes = 2 digits
zone = (`+' | `-') 4 digits
```

`"GET /index.php HTTP/1.1"`
: The line of the client's request is enclosed in double quotes. It contains: the method used by the client here it is `GET`, the requested resource `/index.php` and the protocol used `HTTP/1.1`.

`302`
: This is the status code that the server returns to the client.

`495`
: Indicates the size of the object returned to the client, excluding headers. If no content was returned to the client, this part will contain "`-`".

`"-"`
: Indicates the site from which the client claims to have launched the request. (here "`-`" means no indication).

`"Mozilla/5.0 (iPhone; CPU iPhone OS 7_0 like Mac OS X) AppleWebKit/537.51.1"`
: The User-Agent header of the HTTP request. This information identifies the client browser.

`access.log` file example

The `access_static.log` file listed, below, corresponds to an anthology of the original `/var/log/apache2/access.log` file. This is the file that was processed to generate the result of the static version (*cf.* statique).

```
151.80.39.160 - - [01/Jan/2018:00:19:14 +0100] "GET /claroline/document/
document.php?
cmd=exChDir&file=L0V4ZW1wbGVzL0pBVkEtRUVfV0VCL01WQzItSGliZXJuYXRlL2FjdGlvbn
M%3D&cidReset=true&cidReq=imp_CLD&sort=date&dir=4 HTTP/1.1" 302 722 "-"
"Mozilla/5.0 (compatible; AhrefsBot/5.2; +http://ahrefs.com/robot/)"
151.80.39.105 - - [01/Jan/2018:00:46:42 +0100] "GET /claroline/user/
userInfo.php?uInfo=60155&cidReset=true&cidReq=M3206 HTTP/1.1" 302 558 "-"
"Mozilla/5.0 (compatible; AhrefsBot/5.2; +http://ahrefs.com/robot/)"
151.80.39.162 - - [01/Jan/2018:00:53:49 +0100] "GET /claroline/document/
document.php?
cmd=exChDir&file=L0VKQi9leGVtcGxlcy9wcm9tb3Rpb24yMDEwL3ByZW1pZXItRUpC&cidRe
set=true&cidReq=imp_UML_J2EE&sort=date&dir=4 HTTP/1.1" 302 707 "-"
"Mozilla/5.0 (compatible; AhrefsBot/5.2; +http://ahrefs.com/robot/)"
5.196.87.40 - - [01/Jan/2018:01:00:33 +0100] "GET /claroline/document/
document.php?
cmd=exChDir&file=L0FMVEVSTkFOQ0UvQ2FybmV0c19kZV9saWFpc29u&cidReset=true&cid
Req=ADMINRT&sort=path&dir=3 HTTP/1.1" 302 675 "-" "Mozilla/5.0 (compatible;
AhrefsBot/5.2; +http://ahrefs.com/robot/)" 5.196.87.36 - - [01/Jan/
2018:01:30:20 +0100] "GET /claroline/document/document.php?
cmd=exChDir&file=L1RyYXZhdXhfcHJhdGlxdWVzL0NvbmNlcHRpb24vc29sdXRpb25z&cidRe
set=true&cidReq=imp_BDM1&sort=path&dir=3 HTTP/1.1" 302 698 "-" "Mozilla/5.0
(compatible; AhrefsBot/5.2; +http://ahrefs.com/robot/)" 5.196.87.24 - -
[01/Jan/2018:02:41:39 +0100] "GET /claroline/document/document.php?
cmd=exChDir&file=L0V4ZW1wbGVzL0oyRUVfRUpCMi9FSkIyLVNlc3Npb25TdGF0ZWxlc3M%3D
&cidReset=true&cidReq=imp_CLD&sort=path&dir=3 HTTP/1.1" 302 713 "-"
"Mozilla/5.0 (compatible; AhrefsBot/5.2; +http://ahrefs.com/robot/)"
111.202.100.115 - - [01/Jan/2018:02:46:02 +0100] "GET /robots.txt HTTP/1.1"
302 478 "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/help/
webmasters.htm#07)" 111.202.100.115 - - [01/Jan/2018:02:46:31 +0100] "GET /
HTTP/1.1" 302 457 "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/
- help/webmasters.htm#07)" 54.36.149.91 - - [01/Jan/2018:03:17:24 +0100] "GET
/robots.txt HTTP/1.1" 302 457 "-" "Mozilla/5.0 (compatible; AhrefsBot/5.2;
+http://ahrefs.com/robot/)" 5.50.238.70 - - [29/Aug/2018:17:01:34 +0200]
"GET / HTTP/1.1" 302 495 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/46.0.2486.0 Safari/537.36
Edge/13.10586" 195.220.133.1 - - [29/Aug/2018:17:36:17 +0200] "GET / HTTP/
1.1" 302 495 "-" "Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/537.36
(KHTML, like Gecko) Chrome/52.0.2743.116 Safari/537.36" 180.76.15.154 - -
[29/Aug/2018:17:40:07 +0200] "GET / HTTP/1.1" 302 458 "-" "Mozilla/5.0
```

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Les "traits d'union" indiquent que la portion d'information correspondante n'est pas disponible.

CGI-BIN

Common Gateway Interface

The Common Gateway Interface, usually abbreviated to CGI, is an interface used by web servers.

How it works?

Instead of sending the content of a file (HTML file, image), the web server executes a program and returns the generated content. CGI is the standard that indicates how the request is transmitted between the program (CGI-BIN) and the server (HTTP).

One of the characteristics of the CGI interface is that it relies on standard streams (like `stdin`) to communicate with the server. It is therefore independent of any programming language. Thus, it is possible to write a CGI program in C. The program will only need to make some displays (by simple `printf()`) which will be interpreted by the HTTP server.

From the point of view of the HTTP server, it is necessary to configure it to associate the execution of the CGI program with certain URLs (this configuration has already been done for the apache server that is installed with the virtual machine that is provided to you). Typically, we configure the server so that all the files that are in a certain folder on the server (it is usually called `cgi-bin`) are executed when a request is made to access this file. The result generated by this call can be interpreted by the server.

In a practical way

All types of information that can be viewed by a web browser can be generated (HTML page, GIF image, JPEG, *etc*). You just have to tell the browser what type of information it is. In the case of an HTML page, the program simply displays (with a `printf()`) the first line (note the two line breaks symbolized by the ↲):

```
Content-type: text/html ↲
↲
```

Then any text displayed (still using `printf()`) will be interpreted as HTML code by the browser. You just have to put the executable program in the `/usr/lib/cgi-bin/` directory of the server to be able to visualize its content like any other web page.

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Static version

Sample code

This section gives you the listing of a program in C language which, once compiled and installed, generates the displays corresponding to the analysis of the `access_statique.log` file that can be viewed on the logs page.

This is a static version (the execution of the program always generates the same result whatever the content of the log file).

```c
/*
 * versionstatique.c
 *
 * © Copyright 2024 E. Huot <etienne.huot@uvsq.fr>
 *
 * This program is can only be used for education purpose.
 * It is part of a project, so-called SAÉ, aiming to learn
 * programmation bases using the C language.
 * It can only be used by the students of the RT departement,
 * Technological Institute of Velizy - Université de Versailles - Saint-
 * Quentin-en-Yvelines - Université de Paris-Saclay.
 *
 * The aim of the project is to propose a tool to vizualize a bunch of
 * statistics on a web server logs.
 * The following code propose a static version of the code that have to
 * be developped by the students.
 *
 */


#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <libgraphique.h>

void lecture_style()
{
  /* We could have used a CSS file, but it is sometimes     */
  /* more practical to define the style within the HTML     */
  /* code generated by this CGI-BIN.                        */
  /* We read this style from the file "aecrire.html"        */
  /* and then display it.                                   */
  /*                                                        */
  /* An attentive student will easily notice that this      */
  /* function may not be useless for their own needs...     */

  FILE *fd;
  char str[1024];
  if ((fd = fopen("aecrire.html","r")) == NULL)
    fprintf(stderr,"ERREUR DE CHARGEMENT DE FICHIER\n");
  else {
    while (!feof(fd)){
```

```
      fgets(str,1024,fd);
      printf("%s",str);

    }
    fclose(fd);
  }
}

void versionweb()
{
  printf("Content-type: text/html\n\n");

  printf("<!doctype html>\n");
  printf("<html lang=\"fr\">\n");
  printf("<body>\n");

  lecture_style();

/*
 * Note that for the web, accented characters have been
 * transformed because C character strings are natively
 * encoded in ASCII and not in UTF-8.
 * A student eager to do things properly should normally
 * preprocess all accents before displaying them in HTML.

 */
  printf("Depuis le 01/Jan/2018 on a enregistr&eacute; 998
connexions.<br>\n");

  printf("<div id = \"vertgraph\">\n");
  printf("    <ul>\n");
/*
 * Each bar of the histogram is placed every 31 pixels.
 * The label here corresponds to the height/10.
 * Labels for heights < 25, are not displayed as it would not fit into the
bar
 */
  printf("        <li style=\"left: 10px;  height: 79px;\">8</li>\n");
  printf("        <li style=\"left: 41px;  height: 81px;\">8</li>\n");
  printf("        <li style=\"left: 72px;  height: 99px;\">10</li>\n");
  printf("        <li style=\"left: 103px;  height: 23px;\"></li>\n");
  printf("        <li style=\"left: 134px;  height: 137px;\">14</li>\n");
  printf("        <li style=\"left: 165px;  height: 113px;\">11</li>\n");
  printf("        <li style=\"left: 196px;  height: 127px;\">13</li>\n");
  printf("        <li style=\"left: 227px;  height: 73px;\">7</li> \n");
  printf("        <li style=\"left: 258px;  height: 134px;\">13</li> \n");
  printf("        <li style=\"left: 289px;  height: 12px;\"></li> \n");
  printf("        <li style=\"left: 320px;  height: 115px;\">11</li>\n");
  printf("        <li style=\"left: 351px;  height: 5px;\"></li> \n");
  printf("    </ul>\n");
  printf("</div>\n");

  printf("</html>\n");
  printf("</body>\n\n");

}

void versiontexte()
```

```
{
/*
 * We avoid accented letters in the shell
 */

  printf("Depuis le 01/Jan/2018 on a enregistre 998 connexions.\n");

  printf("   Janvier :  7.9%%\n");
  printf("   Fevrier :  8.1%%\n");
  printf("      Mars :  9.9%%\n");
  printf("     Avril :  2.3%%\n");
  printf("       Mai : 13.7%%\n");
  printf("      Juin : 11.3%%\n");
  printf("   Juillet : 12.7%%\n");
  printf("      Aout :  7.3%%\n");
  printf(" Septembre : 13.4%%\n");
  printf("   Octobre :  1.2%%\n");
  printf("  Novembre : 11.5%%\n");
  printf("  Decembre :  0.5%%\n");
}

void versiongraphique() {
  Point p = {30,50};
  ouvrir_fenetre(430,350);
  dessiner_rectangle((Point){0,0},430,350,blanc);


  afficher_texte("Depuis le 01/Jan/2018 on a enregistré 998 connexions.",12,p

  p.y=300;

afficher_texte(" Jan  Fev  Mar  Avr  Mai  Juin  Juil  Aou  Sep  Oct  Nov
Dec",14,p,vert);

  dessiner_rectangle((Point){30,221},30,79,bleu);
  dessiner_rectangle((Point){61,219},30,81,bleu);
  dessiner_rectangle((Point){92,201},30,99,bleu);
  dessiner_rectangle((Point){123,277},30,23,bleu);
  dessiner_rectangle((Point){154,163},30,137,bleu);
  dessiner_rectangle((Point){185,187},30,113,bleu);
  dessiner_rectangle((Point){216,173},30,127,bleu);
  dessiner_rectangle((Point){247,227},30,73,bleu);
  dessiner_rectangle((Point){278,166},30,134,bleu);
  dessiner_rectangle((Point){309,288},30,12,bleu);
  dessiner_rectangle((Point){340,185},30,115,bleu);
  dessiner_rectangle((Point){371,295},30,5,bleu);

  afficher_texte("8",14,(Point){40,231},blanc);
  afficher_texte("8",14,(Point){71,229},blanc);
  afficher_texte("10",14,(Point){97,211},blanc);

  afficher_texte("14",14,(Point){159,173},blanc);
  afficher_texte("11",14,(Point){190,197},blanc);
  afficher_texte("13",14,(Point){221,183},blanc);
  afficher_texte("7",14,(Point){257,237},blanc);
  afficher_texte("13",14,(Point){283,176},blanc);

  afficher_texte("11",14,(Point){345,195},blanc);
```

```
  actualiser();
  attendre_clic();
  fermer_fenetre();
}


int main(int argc, char* argv[])
{
    if (argc > 1) {
      if (strcmp(argv[1],"-txt")==0) {
        versiontexte();
      }
      if (strcmp(argv[1],"-gr")==0) {
        versiongraphique();
      }

    } else {
      versionweb();
    }

    return EXIT_SUCCESS;
}
```

Running the static web version

Once compiled, the code listed in the previous paragraph, the program must be placed in `/usr/lib/cgi-bin/`.

The figure below shows the execution of the program `/usr/lib/cgi-bin/versionstatique` It is used here with its default interface (web).

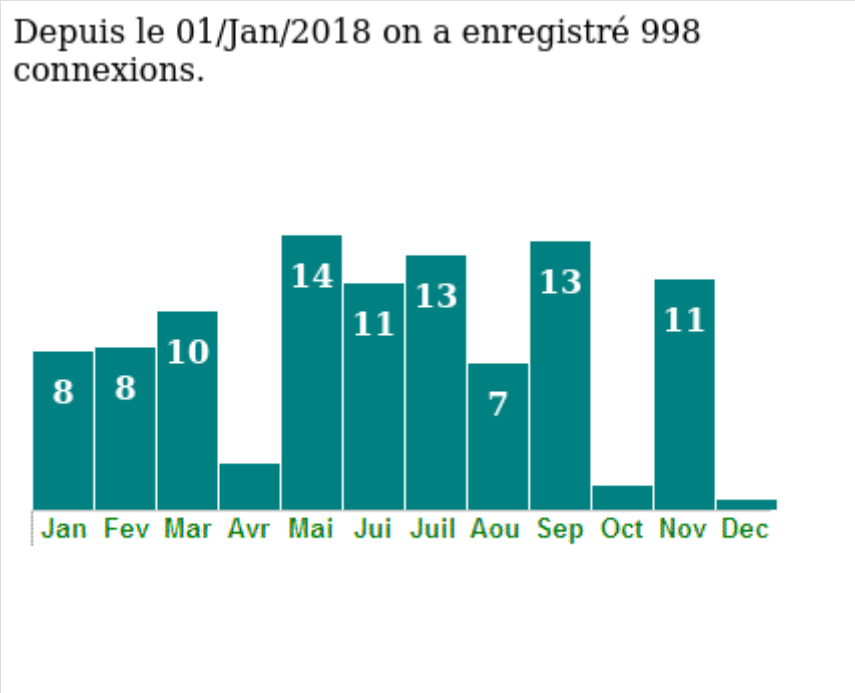


Figure 4: Execution of the static version (web display).

Running the static graphical version

When executed with the `-gr` option as in the following command line:

```
etudiant@virtualdebian:~$ /usr/lib/cgi-bin/versionstatique -gr
```

one obtain a visualization with the graphical interface which relies on the use of the graphical library (`libgraphique`) that you will use in further TP. Note that this version of the library has been slightly modified to use a font installed in a system directory of this Linux distribution and not the font `verdana.ttf` installed in a local directory.

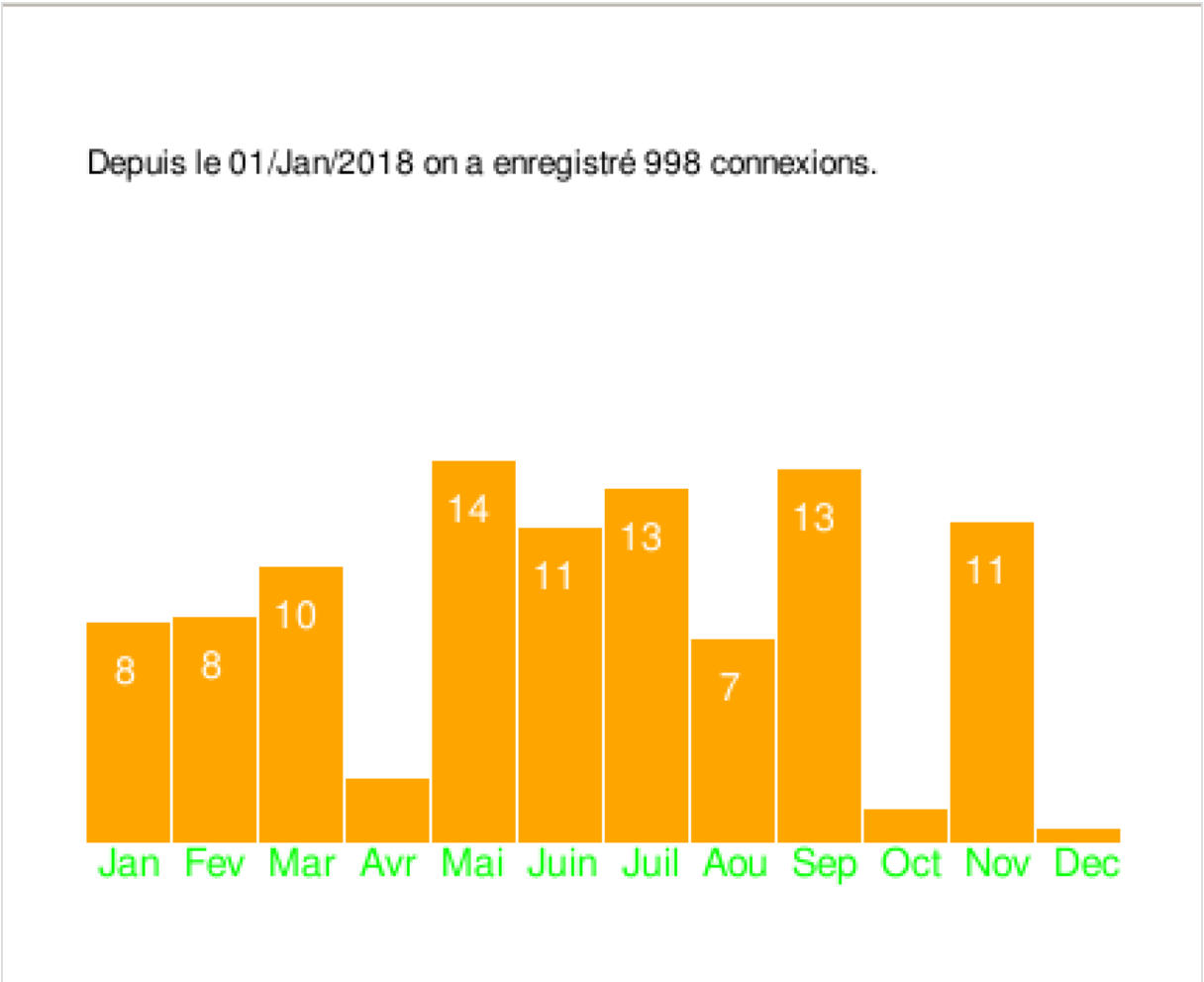


Figure 5: Execution of the static version (graphical display).

Running the static text version

When executed with the `-txt` option as in the following command line:

```
etudiant@virtualdebian:~$ /usr/lib/cgi-bin/versionstatique -txt
```

one obtains the following result:

```
Depuis le 01/Jan/2018 on a enregistre 998 connexions.
   Janvier :  7.9%
   Fevrier :  8.1%
      Mars :  9.9%
     Avril :  2.3%
       Mai : 13.7%
      Juin : 11.3%
   Juillet : 12.7%
      Aout :  7.3%
 Septembre : 13.4%
   Octobre :  1.2%
  Novembre : 11.5%
  Decembre :  0.5%
```

Figure 6: Execution of the static version (graphical display).

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Submission

Deadline

For this project you will work in binomial constituted within the same TP group (possibly in trinomial if odd TP group, **nobody is allowed to work alone**).

You have until January 17th 2027 23:59 to submit your work on eCampus. Do not wait until the last minute. The server may crash, especially if it is saturated with your submissions.

At each submission, check that the upload has been done correctly by downloading your file and checking its integrity.

Archive

Your archive file in `.tgz` format (containing all your files to be submitted - see subject) must imperatively bear the family names (unaccented and without spaces) of the two students of the binomial in alphabetical order and use only one capital letter at the beginning of each family name. For example, for the binomial composed of Tuyet Touchais and Etienne Huot, the file will be called: `HuotTouchais.tgz`. In case of a family name with spaces, for instance, if the binomial is composed of Amar Ramdane Cherif and Etienne Huot, the file name shall be called: `HuotRamdanecherif.tgz`.

Only the `tar` command (not a graphical program like TheUnarchiver, Winzip, xarchiver, etc) should be used to create it.

If your files are correctly placed in the directory `~etudiant/RT1SA05` you just will have to do:

```
etudiant@virtualdebian:~ cd
etudiant@virtualdebian:~ tar cfz HuotTouchais.tgz RT1SA05
```

Upload your file on eCampus

The submission will be done on eCampus in the SAE prog_(RT1SA05) lecture space in the ad hoc section.

Only the first student, in alphabetical order, should submit the work of the binomial, i.e. the `.tgz` archive file.

E. Huot

Bibliography

References & Resources

1. E. Huot. Fondamentaux de la programmation. Langage C. Polycopié de la ressource R107. IUT de Vélizy, Université de Versailles - Saint-Quentin-en-Yvelines, Université Paris-Saclay.
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3. B. Kernighan, D. Ritchie. Le langage C - 2e éd - Norme ANSI. Dunod, 2014.
4. Wikipedia. make.
5. Wikipedia. Log file.
6. Apache HTTP Server. Log Files Documentation.
7. Wikipedia. Common Log Format (CLF).
8. Wikipedia. Common Gateway Interface (CGI).
9. SuperSonique studio. Introduction au CGI (Common Gateway Interface).
10. I. Cohen. CGI/Perl et JavaScript, la création de pages HTML interactives. Eyrolles, pp.294, 1996.
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